

Instructions: Use proper notation, and show all of your work to receive credit. Round numeric answers to at least 2 decimal places.

1. (2 pts) Which of the following is not a property of the cdf, $F(y)$?

- (a) $\lim_{y \rightarrow -\infty} F(y) = 0$.
- (b) $\lim_{y \rightarrow \infty} F(y) = 1$.
- (c) $F(y)$ is nondecreasing.
- (d) $F(y)$ is left-continuous.

2. (2 pts) True or False (circle your answer): The chi-square and exponential distributions are two common types of the gamma distribution.

True

False

3. (3 pts) Suppose that Y has density function

$$f(y) = \begin{cases} ky(1-y), & 0 \leq y \leq 1 \\ 0, & \text{elsewhere} \end{cases}.$$

Find the value of k that makes $f(y)$ a probability density function.

4. (3 pts) If Z is a standard normal random variable, find the value of z_0 such that $P(-z_0 < Z < z_0) = .99$.

5. (5 pts) A supplier of kerosene has a 150-gallon tank that is filled at the beginning of each week. His weekly demand shows a relative frequency behavior that increases steadily up to 100 gallons and then levels off between 100 and 150 gallons. If Y denotes weekly demand in hundreds of gallons, the relative frequency of demand can be modeled by

$$f(y) = \begin{cases} y, & 0 \leq y \leq 1 \\ 1, & 1 < y \leq 1.5 \\ 0, & \text{elsewhere} \end{cases}.$$

Find $F(y)$.

6. (3 pts) Suppose $Y \sim \exp(10)$. Find $P(Y < 3)$. You may use the fact that the cdf of Y is

$$F(y) = \begin{cases} 0, & y < 0 \\ 1 - e^{-y/10}, & y \geq 0 \end{cases}.$$

7. (4 pts) Suppose $X \sim U(a, b)$. Prove that $E(X) = \frac{a+b}{2}$.

8. (3 pts) The fraction of small businesses that show an annual profit in a given year is a random variable whose probability density function is given by

$$f(y) = \begin{cases} ky^2(1-y), & 0 \leq y \leq 1 \\ 0, & \text{elsewhere} \end{cases}.$$

What is the distribution of Y ?